

**VEL TECH RANGARAJAN Dr. SAGUNTHALA R&D INSTITUTE OF
SCIENCE AND TECHNOLOGY**

FRESHMAN ENGINEERING

CDIO WORKSPACE - 1

E-bike Speed Controller System

Theme: PUBLIC SAFETY

ABSTRACT

E-bike Speed Controller System

E-bikes are an innovative solution for affordable and clean energy-based transportation, reducing reliance on fossil fuels. This project focuses on developing an E-bike Motor Speed Controller using an Arduino Uno microcontroller to efficiently regulate motor speed and optimize battery usage. The micro-controller processes throttle inputs, adjusts motor power, and displays speed values on an OLED display, ensuring smooth acceleration and energy efficiency. By integrating a throttle position sensor (TPS) and speed sensor, the system enhances real-time performance monitoring with reduced power consumption. This project aligns with the Affordable and Clean Energy theme by promoting sustainable, cost-effective, and energy-efficient mobility solutions.

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